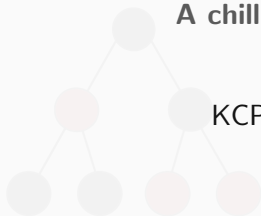


Logic & Lattes

A chill programming social!

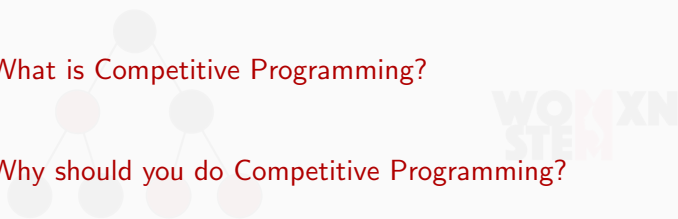


KCPC & KCL WiSTEM

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STEM

March 15, 2026

Overview

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 - 3 Why should you do Competitive Programming?
 - 4 Let's get onto the problems!

KCPC Socials



Instagram



Discord



WhatsApp



Who is KCPC?

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Who is KCPC?

Who is KCPC?

The **King's Competitive Programming Club (KCPC)** is an international student-led organisation based at King's College London dedicated to:

- Algorithmic problem solving
- Programming contests
- Computational reasoning

Our goal: To make competitive programming accessible to students from all backgrounds.

This event is organised jointly with **KCL WiSTEM** to introduce logical reasoning and algorithmic thinking in a relaxed and collaborative setting.

What is Competitive Programming?



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What is Competitive Programming?

Competitive Programming - An Introduction

Competitive programming is a problem-solving discipline where participants write programs to solve algorithmic problems under time constraints.

Problems commonly involve topics such as:

- logical reasoning
- combinatorics and number theory
- graph algorithms
- dynamic programming
- optimisation and data structures

In a typical contest:

- Participants receive a set of algorithmic problems
- Programs must produce correct answers for all valid inputs
- Solutions are evaluated automatically by a judging system

What is Competitive Programming?

Competitive Programming – An Introduction

Competitive programming is not only about writing code.

At its core, it teaches you how to:

- break complex problems into manageable pieces
- reason about constraints and structure
- design efficient algorithms

These skills appear across computer science, mathematics, data science, and many quantitative fields.



Why should you do Competitive Programming?

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Why should you do Competitive Programming?

Why Should You Do Competitive Programming?

Competitive programming develops skills that are valuable both academically and professionally.

- **Structured problem solving** Turning informal ideas into precise algorithms.
- **Algorithmic thinking** Designing efficient solutions instead of brute-force approaches.
- **Programming fluency** Improving coding speed, debugging, and implementation skills.
- **Collaboration** Team contests such as ICPC require cooperative reasoning.

Why should you do Competitive Programming?

Real-World Applications

These skills are widely used in areas such as:

- Software engineering
- Quantitative finance
- Data science and machine learning
- Algorithm research
- Systems engineering

Most importantly, competitive programming is an enjoyable way to explore challenging puzzles and mathematical ideas.

Let's get onto the problems!



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Let's get onto the problems!

Problem Sheet Overview

The problems in this sheet are designed to introduce a range of logical and algorithmic ideas.

They progress roughly in difficulty:

- **Section I:** Logical reasoning and puzzle-style problems
- **Section II:** Algorithmic thinking and programming challenges
- **Section III:** Stretch problems for additional challenge

Let's get onto the problems!

How to Approach the Problems

- Some problems can be solved entirely through reasoning.
- Others may benefit from writing a short program.
- You are encouraged to work individually or in groups.

Discussing different approaches is often one of the best ways to understand a problem.